PRE-PROPOSAL TITLE

Team Alias:	Dutchess
Project Title:	CampusSphere: Role-Based Online Portal with Progressive Web App for Campus Mapping
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Applicable UN Sustainable Development Goals:	Goal 4: Quality Education – Provides accessible and interactive campus navigation for students and visitors. Goal 9: Industry, Innovation, and Infrastructure – Promotes innovation through immersive mapping and scalable web technologies.
Applicable Research Agenda and Priority Area:	ICT for Education – Development of digital platforms to enhance campus accessibility and student experience. Smart Campus Solutions –Integration of interactive mapping and secure authentication for efficient navigation.
Problem Statement:	Camarines Sur Polytechnic Colleges currently lacks an interactive and centralized platform for campus navigation. New students, visitors, and even faculty often face difficulties locating buildings and facilities, relying on static maps or verbal directions. This results in inefficiencies, confusion, and poor user experience. Furthermore, there is no unified system that combines role-based access, secure authentication, and immersive campus mapping in a mobile-friendly format.
Proposed Solution:	The proposed solution is a role-based online portal developed using Node.js and Express.js, designed as a Progressive Web Application (PWA) for mobile-like usability. It will feature: • Secure Google OAuth Authentication restricted to `@cspc.edu.ph` accounts for official users. • Role-based dashboards for Students (with and without CSPC email), Instructors, Admin, and Guests. • Interactive Campus Map with clickable buildings, detailed information, and search functionality. • News and Information Dashboard for announcements and updates. • Scalable architecture to allow future integration of features such as AR navigation, event mapping, and offline mode.
Scope of the Proposed Solution:	• Student Role (with CSPC email): Access personal info, enrollment details, news dashboard, and interactive campus map. • Student Role (without CSPC email): Limited access to campus map, news, and general information. • Instructor Role: Manage announcements, view assigned subjects, and update profile information. • Admin Role: Manage users, update news and FAQs, monitor system logs.

	• Guest Role: Access campus map, school news, and contact details. • Interactive Campus Map: Provides immersive navigation with building details and search functionality.
Limitations of the Proposed Solution:	• Real-time GPS navigation will not be included in the initial version but can be added later. • Integration with existing CSPC systems (e.g., SIAS) may be simulated due to restricted access. • Offline mode and AR features will be considered for future upgrades.
Software Development Methodology (SDM):	Agile Methodology will be used to ensure flexibility and iterative development. The project will follow these phases: 1. Requirement Analysis – Gather functional and non-functional requirements. 2. Design – Create system architecture, ERD, and UI mockups. 3. Development – Implement backend (Node.js, Express.js), frontend (HTML/CSS/JS or React), and interactive map integration using Three.js or Mapbox. 4. Testing – Conduct unit, integration, and user acceptance testing. 5. Deployment – Host on a cloud platform (e.g., Firebase or AWS). 6. Maintenance & Scalability – Prepare for future upgrades and feature enhancements.

☑ Free from duplication from prior studies ☐ Contains duplication with prior studies

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